Beyond the Bell Curve: From Statistical Fit to Investment Decisions

Tianxudong Tang

Abstract

This study examines how different return distributions affect financial risk estimation and portfolio performance. Using daily returns of 20 technology and energy stocks from 2021–2025, I compare Normal, Student-*t*, and two-component Gaussian mixture models (GMM). The models are fitted on 2021–2023 data and evaluated out of sample over 2024–2025. Student-*t* achieves the lowest Bayesian information criterion (BIC) for all 20 stocks, but does not consistently outperform the other models in value-at-risk (VaR) and expected shortfall (ES) tests. In a 502-day portfolio backtest, the fully invested benchmark returns 91.60%, compared with 21.49–24.17% for three VaR-based position-sizing strategies. These strategies reduce maximum drawdown from 20.72% to 6.09–6.78%, but also limit average market exposure to 33.15–36.98%. Their Sharpe ratios remain below that of the benchmark. A supplementary European call option pricing exercise compares theoretical prices under different return distribution assumptions. The results suggest that better historical distribution fitting does not necessarily improve risk forecasts or investment performance. In the rising market studied here, reducing risk came at a substantial opportunity cost. Investors seeking higher returns must also be willing to accept greater risk. No risk, no gain.

Keywords: Portfolio risk management, Opportunity cost, Market exposure, Student-*t*, Value-at-risk, Expected shortfall

1. Introduction

Risk management is an important part of portfolio investment. Investors use statistical models to estimate potential losses and adjust their positions accordingly. Previous studies have shown that volatility-based portfolio management can improve investment performance under certain conditions [1, 2]. However, reducing market exposure does not only limit potential losses. It also limits potential gains when the market moves upward. Therefore, it is worth examining whether a more conservative risk management strategy actually produces a better investment outcome.

In this project, I investigate three related questions. First, how well do different statistical distributions describe historical stock returns, particularly extreme returns? Second, does a model that fits historical data better also provide more accurate risk estimates for future observations? Third, when these risk estimates are used to determine portfolio exposure, how do they affect investment returns and drawdowns?

To address these questions, I analyze 20 US-listed stocks from the technology and energy sectors. The technology group includes MSFT, AAPL, NVDA, GOOGL, META, AMZN, AMD, INTC, ORCL, and CRM. The energy group includes XOM, CVX, COP, EOG, OXY, SLB, HAL, VLO, MPC, and PSX. These groups are based on general business characteristics rather than a strict industry classification. I use data from 2021–2023 to fit three return distributions: Normal, Student-*t*, and a two-component Gaussian mixture model. Their performance is then evaluated using data from 2024–2025.

The Normal distribution is commonly used in financial modeling because of its mathematical simplicity. However, financial returns often exhibit heavier tails than a Normal distribution predicts [3, 4]. The Student-*t* distribution allows for heavier tails, while the Gaussian mixture model provides additional flexibility by combining two Normal components. I compare these models using the Bayesian information criterion (BIC), followed by out-of-sample VaR and ES analysis.

Beyond comparing statistical models, I also examine their practical implications for portfolio management. I construct a fully invested benchmark and three VaR-based position-sizing strategies. The strategies adjust portfolio exposure according to estimated risk, allowing me to compare cumulative returns, volatility, maximum drawdown, and Sharpe ratios. I also include a supplementary theoretical European call option pricing exercise to illustrate another application of return distribution models.

The results show that Student-*t* provides the best historical fit for all 20 stocks, but its advantage in out-of-sample risk estimation is less clear. In the portfolio backtest, all three risk-controlled strategies achieve substantially smaller drawdowns than the fully invested benchmark. However, they also produce much lower cumulative returns and slightly lower Sharpe ratios. This raises an important question about the cost of risk management: how much potential return should an investor be willing to give up to reduce downside risk?

The purpose of this study is not to argue that risk management is unnecessary. Instead, I aim to examine the trade-off between risk reduction and investment performance using a consistent set of historical data and modeling assumptions. The results are specific to the selected stocks, test period, and portfolio strategies, and should not be interpreted as a general prediction of future market performance.

2. Data and Research Design

2.1. Return construction and sample split

Adjusted daily closing prices are collected from January 2021 through December 2025. For stock i, the daily log return is

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \tag{1}$$

Returns are expressed in percentage points when fitting the daily distributions. For the risk analysis, portfolio daily simple returns are expressed as decimal fractions. Keeping these conventions separate avoids an unintended factor-of-100 error.

The training period covers 2021–2023. The 2024–2025 period is reserved for out-of-sample evaluation. Model parameters and fixed risk thresholds are obtained from training data rather than refitted using test-period observations. The test set contains 502 trading days.

For descriptive tail comparisons, standardized returns are

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_i}{s_i}, \tag{2}$$

where $\bar{r}_i$ and s_i are calculated within the period being described. These standardized observations summarize historical extremes rather than prospective probabilities.

2.2. Candidate return distributions

I fit three distributions to each stock's daily log returns by maximum likelihood. The Normal density is

$$f_N(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{3}$$

The location-scale Student-t density is

$$f_t(x) = \frac{1}{s} t_\nu\left(\frac{x-m}{s}\right), \tag{4}$$

where ν controls tail thickness. Smaller ν corresponds to heavier polynomial tails. The two-component Gaussian mixture is

$$f_{\text{GMM}}(x) = w\phi(x;\mu_1,\sigma_1^2) + (1-w)\phi(x;\mu_2,\sigma_2^2). \tag{5}$$

The Normal, Student-t, and GMM specifications have 2, 3, and 5 fitted parameters, respectively. The GMM is a static mixture rather than a time-varying regime model.

To balance fit against parameter count, I use the Bayesian information criterion (BIC) [5],

$$\text{BIC}_M = k_M \log N - 2\sum_{t=1}^{N} \log f_M(r_t). \tag{6}$$

Lower BIC is preferred. For comparisons across stocks, I define

$$\Delta\text{BIC}_{i,M} = \text{BIC}_{i,M} - \min_{M'} \text{BIC}_{i,M'}. \tag{7}$$

The same training observations are used to compare the three distributions for each stock.

3. Statistical Fit: A Clear Winner, but Not Yet an Investment Edge

3.1. Visual comparison and model selection

Figure 1 displays six examples of training-period return histograms and fitted densities. The Student-t curve often follows the sharp central peak more closely than the Normal curve. The models nevertheless differ by stock, and visual inspection alone cannot establish a ranking.

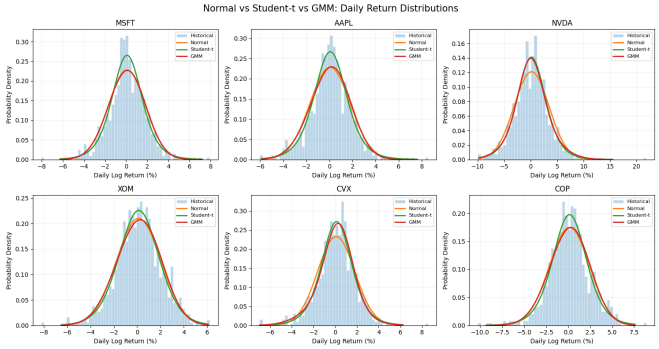


Figure 1: Training-period daily log-return histograms and fitted Normal, Student-t, and two-component GMM densities for six selected stocks, 2021–2023. Returns are expressed in percent. These examples illustrate the fitted shapes and are not a separate test sample.

Student-t achieves the lowest training-period BIC for all 20 stocks. The second-place ranking varies across the two groups. GMM ranks ahead of Normal for eight of the ten technology stocks, while Normal ranks ahead of GMM for seven of the ten energy stocks.

Group	t best	GMM 2nd	Normal 2nd
Technology (10)	10	8	2
Energy (10)	10	3	7
Total (20)	20	11	9

Table 1: Training-period BIC ranking counts across 20 stocks.

The strength of the Student-t advantage is not uniform. For XOM, HAL, and MPC, the BIC differences between Normal and Student-t are only 2.3, 1.7, and 1.7, respectively. For META and ORCL, GMM is substantially closer to Student-t than Normal is. This variation matters when moving from a statistical fit to a financial application. A common winner across stocks does not imply a common economic effect.

3.2. Tail characteristics

Figure 2 reports the fitted Student-t degrees of freedom for all 20 stocks. The technology group has a mean ν of 4.898 and median of 4.558. The corresponding energy values are 8.370 and 8.152. The lower fitted degrees of freedom for technology stocks indicate heavier tails within the chosen Student-t specification.

As a separate descriptive check, I count observations beyond three sample standard deviations. Under a standard Normal

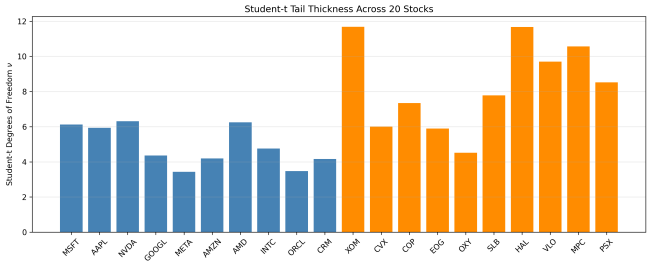


Figure 2: Maximum-likelihood Student-t degrees of freedom for the 20 stocks, fitted on 2021–2023 returns. Smaller ν indicates heavier fitted tails.

benchmark, the probability in either one-sided tail is approximately 0.135%. In the training sample, the technology group's mean frequencies are 0.585% on both sides. The energy group averages 0.532% in the left tail and 0.279% in the right tail. These values are calculated stock by stock and then averaged across each group.

The Student-t degrees of freedom summarize a symmetric fitted distribution. They do not capture every aspect of the observed left-right imbalance. The group comparisons are also descriptive. The stocks share market shocks, so they cannot be treated as 20 independent replications of a sector-level experiment.

4. Out-of-Sample Risk Forecasts: The Advantage Weakens

4.1. VaR and ES evaluation

The next question is whether a model that fits the training returns well also provides useful estimates of future portfolio losses. Tail-risk estimation is especially challenging when financial returns are heteroscedastic and heavy-tailed [6]. The portfolio risk exercise evaluates the same underlying portfolio over the 2024–2025 test period. Model-specific VaR and ES thresholds are estimated from the training period and held fixed during testing. This is a static forecast comparison, not an adaptive volatility model.

Let R_t denote a portfolio daily simple return and $L_t = -R_t$ its loss. For confidence level α, VaR is the α quantile of the loss distribution,

$$\mathrm{VaR}_\alpha = \inf\{l : \Pr(L \le l) \ge \alpha\}. \tag{8}$$

For a continuous loss distribution, expected shortfall is the mean loss conditional on exceeding that quantile,

$$\mathrm{ES}_\alpha = E[L \mid L \ge \mathrm{VaR}_\alpha]. \tag{9}$$

The general quantile-integral definition is used when the distribution is not continuous. Expected shortfall is closely related to the coherent risk-measure framework and conditional value-at-risk optimization [7, 8].

A VaR exceedance occurs when the realized loss exceeds the model's VaR threshold. Under correct unconditional coverage, the expected number of exceedances across N days is $N(1-\alpha)$. Here $N = 502$, giving 25.10 expected exceedances at 95% confidence and 5.02 at 99%.

4.2. Exceedance counts and realized losses

Figure 3 compares observed exceedance counts with their nominal expectations. At 95% confidence, Normal and GMM each record 12 exceedances and Student-t records 14. All three counts are below the expected 25.10. The fixed thresholds therefore appear conservative over this particular test period.

At 99% confidence, Normal records six exceedances, Student-t four, and GMM five. These are all close to the expected 5.02. With so few extreme observations, a difference of one or two exceedances is not strong evidence that one model is statistically superior. Formal backtesting methods distinguish unconditional coverage from the independence of exceedances [9, 10].

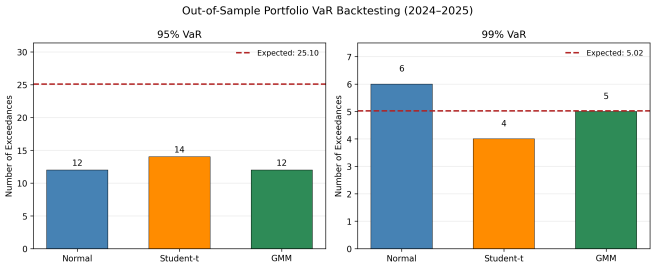


Figure 3: Out-of-sample VaR exceedance counts for 502 portfolio trading days in 2024–2025. Dashed horizontal lines indicate expected counts under nominal 95% and 99% coverage.

Figure 4 places the three fixed 99% VaR thresholds alongside realized daily portfolio returns. The predicted loss thresholds are approximately 3.86% for Normal, 4.30% for Student-t, and 3.96% for GMM. The largest negative observations are concentrated in a small number of trading days, including a pronounced episode in spring 2025. The plot suggests that the volatility environment changes over time, although clustering should be evaluated formally rather than inferred from a picture alone. Conditional heteroscedasticity provides a well-established framework for studying such variation [11].

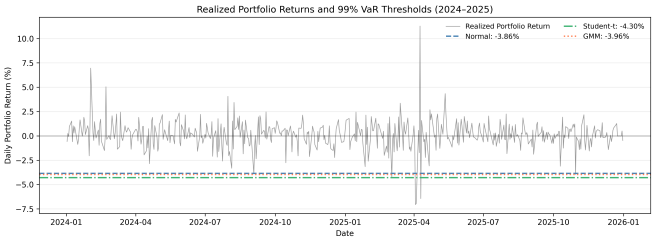


Figure 4: Realized daily simple portfolio returns in 2024–2025 and fixed, training-based 99% VaR return thresholds. A return below a model's horizontal threshold is a VaR exceedance.

4.3. Predicted versus realized VaR and ES

Figure 5 compares model-based loss estimates with empirical summaries of the test-period loss distribution. The realized 95% VaR is approximately 1.95%, compared with model estimates of 2.58–2.74%. The realized 95% ES is 3.15%, compared with model estimates of 3.40–3.68%.

At 99% confidence, the realized VaR is approximately 3.91%. GMM is closest at 3.96%, followed by Normal at 3.86% and Student-t at 4.30%. The realized 99% ES is approximately 5.45%. Student-t is closest at 5.60%, while GMM and Normal give lower estimates of 4.57% and 4.44%.

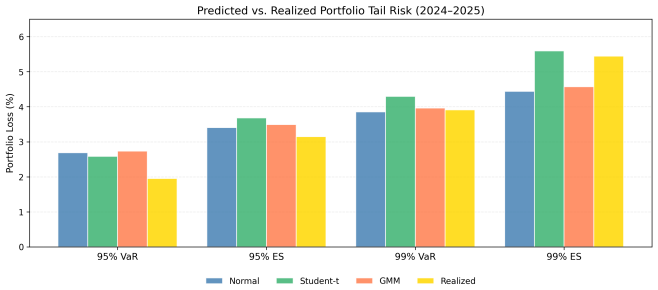


Figure 5: Model-predicted portfolio VaR and ES compared with empirical test-period loss quantiles and tail means, 2024–2025. Values are percentages of portfolio value. Realized values are retrospective summaries rather than forecasts.

	95% VaR	95% ES	99% VaR	99% ES
Normal	2.685	3.403	3.856	4.439
Student-t	2.581	3.683	4.298	5.595
GMM	2.736	3.489	3.965	4.572
Realized	1.950	3.147	3.913	5.448

Table 2: Model-based portfolio tail-risk estimates and empirical test-period benchmarks, in percent.

These results do not identify one universally best model. Student-t provides the closest 95% VaR and 99% ES estimates, while GMM is closest to the realized 99% VaR and Normal is closest to the realized 95% ES. The empirical tail estimates are themselves noisy, especially at 99% confidence, where only a handful of days determine the outcome. A formal comparison would require coverage and exceedance-independence tests, alongside an evaluation of the statistical uncertainty in ES estimates.

5. The Investment Test: The Opportunity Cost of Protection

5.1. Position-sizing rule and implementation

The preceding results compare the statistical quality of fixed risk thresholds. A separate backtest asks what happens when those estimates guide an investment decision. The economic question is related to earlier studies of volatility timing and volatility-managed portfolios [1, 2]. I use the same underlying portfolio in all four strategies and compare a fully invested benchmark with Normal-, Student-t-, and GMM-based position sizing. The backtest covers the 502 trading days in 2024–2025.

The fitted 1% portfolio return quantile q_M for each model M comes from the 2021–2023 training sample. During the test period, I scale the distance between that quantile and the fitted model center m_M by a volatility multiplier,

$$v_t = \mathrm{SD}(R_{t-63}, \ldots, R_{t-1}), \tag{10}$$

$$h_t = \mathrm{clip}\left(\frac{v_t}{s_{\mathrm{train}}}, 0.25, 4\right), \tag{11}$$

$$q_{M,t} = m_M + h_t(q_M - m_M), \tag{12}$$

$$\widehat{\mathrm{VaR}}^{99\%}_{M,t} = \max(-q_{M,t}, 10^{-8}), \tag{13}$$

$$a_{M,t} = \min\left(1, \frac{0.01}{\widehat{\mathrm{VaR}}^{99\%}_{M,t}}\right). \tag{14}$$

Here s_{train} is the training-period standard deviation of the portfolio's simple returns, and $a_{M,t}$ is the fraction of capital invested in the underlying portfolio. The rolling standard deviation uses only returns through day $t-1$, so the exposure for day t does not use that day's realized return. The maximum exposure is 100%, with no leverage. The remaining capital earns zero in this experiment. The benchmark holds $a_t = 1$ throughout.

Transaction costs are estimated at five basis points per unit of absolute exposure change. The daily net strategy return and accumulated wealth are

$$R^{\mathrm{net}}_{M,t} = a_{M,t}R_t - 0.0005\,|a_{M,t} - a_{M,t-1}|, \tag{15}$$

$$W_{M,t} = \prod_{u \le t}(1 + R^{\mathrm{net}}_{M,u}). \tag{16}$$

The initial move from cash to the chosen exposure is charged as turnover. This is a dynamic volatility-scaled strategy, not a rolling re-estimation of the three return distributions. It also differs from the static VaR backtest in the preceding section.

5.2. Wealth trajectories, foregone gains, and performance

Figure 6 shows the resulting net wealth paths. The fully invested portfolio grows from 1 to 1.916, corresponding to a cumulative return of 91.60%. Its path is considerably more volatile, including a sharp drawdown around spring 2025. All three VaR-controlled portfolios follow much smoother paths and end between 1.215 and 1.242. Their limited participation in the underlying portfolio's rally is the visible cost of reducing exposure.

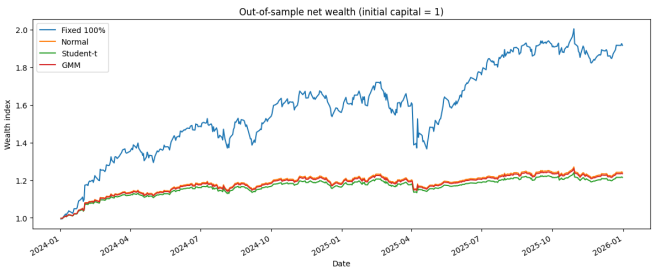


Figure 6: Out-of-sample net wealth of the fully invested portfolio and three volatility-scaled, VaR-based position-sizing strategies, 2024–2025. Initial wealth is one. Returns include estimated exposure-change transaction costs. Cash earns zero.

Strategy	Return (%)	Vol. (%)	Max DD (%)	Sharpe
Fixed 100%	91.60	22.77	-20.72	1.548
Normal	24.17	8.00	-6.78	1.399
Student-*t*	21.49	7.17	-6.09	1.399
GMM	23.45	7.78	-6.60	1.399

Strategy	99% VaR (%)	99% ES (%)	Avg. exposure (%)
Fixed 100%	3.91	5.45	100.00
Normal	1.38	1.80	36.98
Student-*t*	1.24	1.62	33.15
GMM	1.35	1.75	35.96

Table 3: Out-of-sample strategy performance, 2024–2025. VaR and ES are empirical 99% daily loss measures of strategy net returns. Annualized volatility and Sharpe ratios assume zero cash return.

Strategy	Total turnover	Estimated cost
Fixed 100%	1.0000	0.050%
Normal	3.1061	0.155%
Student-*t*	2.7765	0.139%
GMM	3.0158	0.151%

Table 4: Exposure turnover and estimated transaction costs. Turnover is the sum of absolute daily changes in exposure, including the initial investment. Estimated costs are fractions of initial capital under the backtest convention.

5.3. What the investor actually gave up

The headline comparison is not a small statistical difference. Relative to the 91.60% fully invested return, the Normal, Student-*t*, and GMM strategies surrender 67.43, 70.11, and 68.15 percentage points of cumulative return, respectively. Expressed as a fraction of the benchmark's cumulative gain, the forgone gain is approximately 74%, 77%, and 74%. These are descriptive differences in realized backtest outcomes, not estimates of the gains that would be sacrificed in an average future market.

The other side of the trade-off is real. The Student-*t* rule cuts the magnitude of maximum drawdown by 14.63 percentage points, from 20.72% to 6.09%, and reduces annualized volatility by 15.60 percentage points. An investor facing strict loss limits might rationally prefer that smoother path. But an investor whose objective is to maximize growth or the observed Sharpe ratio would not have benefited from these particular risk controls over this sample.

The exposure figures help explain the contrast. The VaR strategies invest only about one-third of capital in equities on average, with the remainder earning zero in the assumed cash account. This mechanical reduction in participation is an important alternative explanation for their lower volatility and smaller drawdowns. The experiment cannot establish how much of the protection comes from successful timing rather than from holding less equity. Nevertheless, it makes the economic trade-off visible: risk was reduced by accepting substantially less market participation.

The risk-controlled strategies lower annualized volatility from 22.77% to 7.17–8.00% and maximum drawdown from 20.72% to 6.09–6.78%. Student-*t* produces the smallest realized losses, but it also has the smallest average market exposure at 33.15%. Its lower risk cannot be attributed to superior risk timing without controlling for this difference in invested capital. The Normal and GMM strategies take slightly more risk and earn slightly higher total returns.

The three model-based strategies have almost indistinguishable annualized Sharpe ratios, around 1.399, below the fully invested benchmark's 1.548. Their near-identical risk-adjusted results are consistent with the fact that they invest in the same underlying portfolio and vary mainly in exposure. The evidence therefore supports a narrower conclusion than "Student-*t* improves investment performance." VaR-based position sizing reduces risk in this sample, but the improvement is accompanied by lower participation in gains, and the historical BIC winner does not deliver a clear advantage in risk-adjusted performance. A matched-exposure or matched-risk-budget experiment would be needed to isolate the incremental value of the distribution model.

5.4. Did model sophistication pay for itself?

The distinction between the three VaR rules is much smaller than the distinction between any VaR rule and full investment. Their annualized Sharpe ratios differ by only about 0.0005 using unrounded values, despite their very different distributional assumptions and Student-*t*'s universal BIC lead. The more elaborate historical fit therefore does not translate into an observable Sharpe improvement within this experiment.

This is not a general argument against downside protection. It is a warning against treating lower realized volatility or a better in-sample likelihood as sufficient evidence of a superior investment rule. The tested risk budget, the no-leverage constraint, the volatility-scaling formula, and the positive returns of the underlying portfolio during 2024–2025 all contribute to the observed opportunity cost. A different market regime or investment objective could reverse the preference. The conclusion supported here is specific and actionable: when judging a risk overlay, report both the protection obtained and the return and market exposure surrendered.

6. Additional Evidence: Distribution Assumptions and Option Values

6.1. Why an unmodified Student-t model is problematic

The strong historical fit of Student-*t* raises a separate question about its use for option valuation. For a one-period log return X,

$$S_{t+\Delta t} = S_t e^X. \tag{17}$$

A risk-neutral stock-price model requires a finite expected growth factor. For Gaussian $X \sim N(\mu, \sigma^2)$,

$$E[e^X] = \exp\left(\mu + \frac{\sigma^2}{2}\right) < \infty. \tag{18}$$

A Student-*t* density instead has a polynomial positive tail, $f_t(x) \sim Cx^{-(\nu+1)}$. The factor e^x dominates this decay, and

$$E[e^X] = \int_{-\infty}^{\infty} e^x f_t(x)\, dx = \infty. \tag{19}$$

Adding a finite drift shift does not fix this divergence. Figure 7 illustrates the different behavior of the exponential-moment integrands.

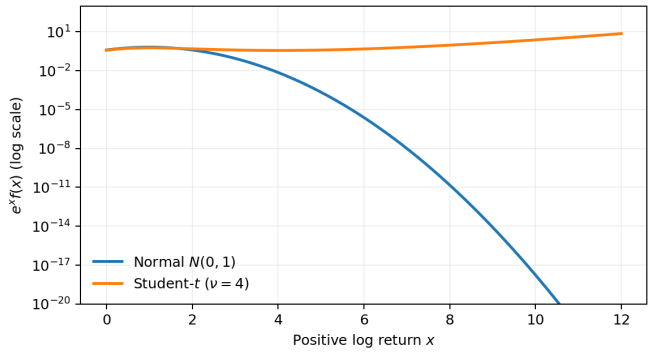


Figure 7: Positive-tail exponential-moment integrand $e^x f(x)$ for standard Normal and Student-t ($\nu = 4$) densities. The logarithmic vertical axis highlights the divergence of the untruncated Student-t exponential moment.

To construct a finite-moment illustration, I truncate the fitted Student-t density to finite bounds L and U,

$$f_{\text{tr}}(x) = \frac{f_t(x)}{F_t(U) - F_t(L)}, \qquad L \le x \le U. \tag{20}$$

The finite exponential moment under the truncated density is

$$E[e^{X^{\text{tr}}}] = \frac{\int_L^U e^x f_t(x)\,dx}{F_t(U) - F_t(L)} < \infty. \tag{21}$$

The risk-neutral log-return increment is then specified as

$$c_{\text{tr}} = \log E[e^{X^{\text{tr}}}], \tag{22}$$

$$R^Q = (r_f - q)\Delta t + X^{\text{tr}} - c_{\text{tr}}. \tag{23}$$

This construction satisfies $E^Q[e^{R^Q}] = e^{(r_f - q)\Delta t}$. It is one modeling choice, not a unique risk-neutral measure implied by historical data. Truncation can materially change the right tail that matters for call payoffs.

6.2. *Option price comparisons*

I compare theoretical European call prices for META and XOM at maturities of 21 and 126 trading days and strike-to-spot ratios of 0.9, 1.0, and 1.1. The benchmark is Black–Scholes [12], alongside Gaussian Monte Carlo, GMM, and truncated Student-t simulations. All model-based call prices are interpreted through

$$C_0 = e^{-r_f T} E^Q[(S_T - K)^+]. \tag{24}$$

Figure 8 shows the normalized call prices.

Gaussian Monte Carlo closely tracks Black–Scholes, as expected when their Gaussian dynamics and parameters are consistent. GMM prices are also generally close to the benchmark.

The truncated Student-t prices are noticeably lower for META, but remain much closer to Black–Scholes for XOM. Figure 9 quantifies the difference as

$$\Delta C = 100 \frac{C_{\text{model}} - C_{\text{BS}}}{C_{\text{BS}}}. \tag{25}$$

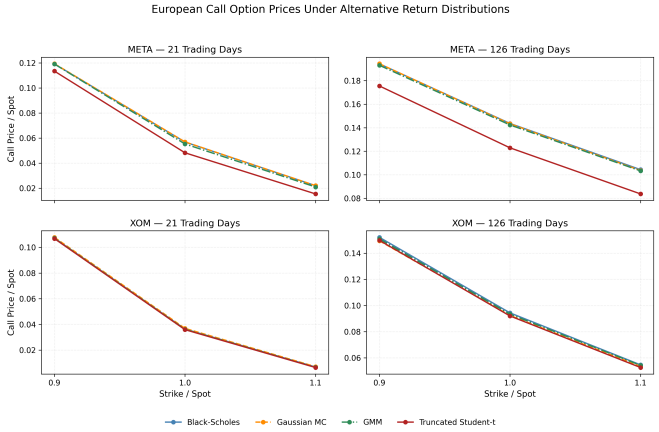


Figure 8: Normalized theoretical European call prices for META and XOM under Black–Scholes, Gaussian Monte Carlo, GMM, and truncated Student-t assumptions. Maturities are 21 and 126 trading days.

For META's 21-day out-of-the-money call, the truncated Student-t price is 29.91% below Black–Scholes. The corresponding difference for XOM is 5.87%. At 126 days, the out-of-the-money differences are 19.75% and 3.75%, respectively.

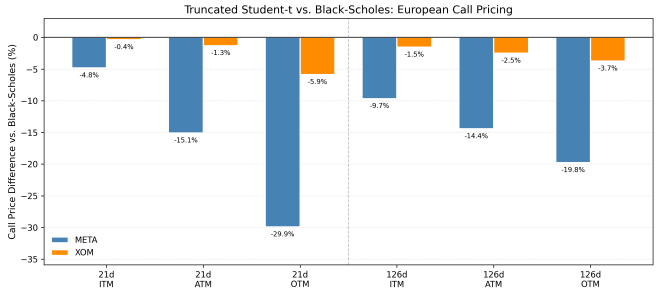


Figure 9: Relative differences between truncated Student-t and Black–Scholes theoretical call prices. Negative values indicate lower prices under the truncated Student-t construction. ITM, ATM, and OTM refer to strike-to-spot ratios of 0.9, 1.0, and 1.1.

These results should not be read as evidence that heavier tails always lower call prices. The fitted center and scale, the finite truncation bounds, and the drift adjustment all affect the simulated right-tail payoffs. Nor do differences from Black–Scholes establish pricing errors against market option quotes. The exercise demonstrates sensitivity to assumptions, not market pricing accuracy.

7. Discussion and Limitations

The study's most important distinction is between protecting capital and improving an investment outcome. Student-t is unambiguously preferred by training-period BIC, but its out-of-sample tail-risk performance is mixed. More importantly, its lower realized strategy risk is accompanied by the lowest average invested fraction. A model that induces a smaller position can generate a smaller drawdown without providing a better prediction of when losses will occur. The nearly identical Sharpe ratios across the three risk overlays reinforce the absence of a clear risk-adjusted advantage in this particular test.

The market setting is essential to the interpretation. Unlike studies finding economic benefits from volatility-based timing rules [1, 2], this particular backtest finds no improvement in the observed Sharpe ratio. These differences should not be interpreted as a direct replication test because the assets, timing rules, and constraints differ. The fully invested portfolio experienced a large cumulative gain over 2024–2025. In such a sample, spending long periods at roughly one-third equity exposure creates a substantial opportunity cost. The result should not be extrapolated into a claim that full investment is optimal in downturns or that it has a higher future probability of profit. No future profit probability is estimated here, and the reported Sharpe ratios are descriptive rather than statistically differences.

The analysis covers 20 selected stocks and one train-test split. The static VaR coverage study uses fixed training thresholds, whereas the separate portfolio strategy applies a lagged 63-day volatility multiplier to training-period distribution quantiles. The strategy outcomes depend on the 1% VaR budget, exposure cap, transaction-cost assumption, and zero return on uninvested cash. The models' different average exposures prevent a causal attribution of strategy performance to distribution quality alone. At 99% confidence, the realized ES estimate also rests on very few extreme observations. The supplementary option prices are theoretical outputs from assumed risk-neutral constructions, not tests against market option quotes. These boundaries define the claims rather than invalidating the observed comparisons.

8. Conclusion

This study compares three statistical return distributions and examines how their risk estimates translate into portfolio investment decisions. The Student-t distribution provides the best historical fit among the three models for all 20 stocks. However, this advantage does not consistently translate into better out-of-sample VaR and ES forecasts. A better statistical fit does not necessarily imply more accurate risk predictions or better financial decisions.

The portfolio backtest reveals a more practical trade-off. During 2024–2025, the fully invested benchmark achieves a cumulative return of 91.60%, compared with 21.49–24.17% for the three VaR-based strategies. Although these strategies reduce maximum drawdown from 20.72% to approximately 6–7%, their average market exposure falls to roughly one-third. They sacrifice 67–70 percentage points of cumulative return, while their Sharpe ratios remain below that of the benchmark. Student-t produces the lowest volatility and drawdown, but also the lowest cumulative return and market exposure.

These findings highlight the limitations of relying on historical distributional fit alone for financial decision-making. While the tested risk controls successfully reduce losses during market declines, they also limit participation when prices rise. In the rising market examined here, lower portfolio risk does not translate into better realized risk-adjusted performance. This does not imply that remaining fully invested is always preferable, nor does it establish that more complex models would necessarily perform better.

Future research could explore more adaptive approaches that account for changing volatility, market regimes, and return distributions. However, any additional complexity must demonstrate its value through rigorous out-of-sample evaluation and improved investment outcomes, rather than statistical fit alone.

Ultimately, risk protection comes at a cost, and investors must decide whether that protection is worth the returns they sacrifice. The objective is not simply to minimize risk, but to balance risk and opportunity. After all, no risk, no gain.

References

[1] J. Fleming, C. Kirby, B. Ostdiek, The economic value of volatility timing, The Journal of Finance 56 (1) (2001) 329–352. doi:10.1111/0022-1082.00327.

[2] A. Moreira, T. Muir, Volatility-managed portfolios, The Journal of Finance 72 (4) (2017) 1611–1644. doi:10.1111/jofi.12513.

[3] E. F. Fama, The behavior of stock-market prices, The Journal of Business 38 (1) (1965) 34–105. doi:10.1086/294743.

[4] R. Cont, Empirical properties of asset returns: Stylized facts and statistical issues, Quantitative Finance 1 (2) (2001) 223–236. doi:10.1080/713665670.

[5] G. Schwarz, Estimating the dimension of a model, The Annals of Statistics 6 (2) (1978) 461–464. doi:10.1214/aos/1176344136.

[6] A. J. McNeil, R. Frey, Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach, Journal of Empirical Finance 7 (3–4) (2000) 271–300. doi:10.1016/S0927-5398(00)00012-8.

[7] P. Artzner, F. Delbaen, J.-M. Eber, D. Heath, Coherent measures of risk, Mathematical Finance 9 (3) (1999) 203–228. doi:10.1111/1467-9965.00068.

[8] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, The Journal of Risk 2 (3) (2000) 21–41. doi:10.21314/JOR.2000.038.

[9] P. H. Kupiec, Techniques for verifying the accuracy of risk measurement models, The Journal of Derivatives 3 (2) (1995) 73–84. doi:10.3905/jod.1995.407942.

[10] P. F. Christoffersen, Evaluating interval forecasts, International Economic Review 39 (4) (1998) 841–862. doi:10.2307/2527341.

[11] R. F. Engle, Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation, Econometrica 50 (4) (1982) 987–1007. doi:10.2307/1912773.

[12] F. Black, M. Scholes, The pricing of options and corporate liabilities, Journal of Political Economy 81 (3) (1973) 637–654. doi:10.1086/260062.